

Parametric Study and Optimization of Wire Electrical Discharge Machining Parameters for SS304 Stainless Steel

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ABSTRACT

Wire Electrical Discharge Machining (WEDM) is a widely used non-traditional machining process for precision machining of electrically conductive materials with complex geometries. Stainless steel SS304 is extensively employed in aerospace, automotive, medical, and gauge-manufacturing industries due to its excellent corrosion resistance, toughness, and biocompatibility; however, its machining using conventional methods is difficult because of work hardening and poor machinability. The present study investigates the influence of major WEDM process parameters pulse on-time, pulse off-time, and peak current on surface roughness during machining of SS304 stainless steel. Experiments were designed using Taguchi's design of experiments approach to reduce experimental effort while maintaining result reliability. Empirical modeling, signal-to-noise ratio analysis, analysis of variance (ANOVA), and desirability-based optimization were employed to identify optimal machining conditions. Confirmation experiments were conducted to validate the predicted results. The study demonstrates that appropriate selection and optimization of WEDM parameters significantly improve surface finish and machining quality of SS304 components.

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1. INTRODUCTION

1. Introduction

Wire Electrical Discharge Machining (WEDM) is an advanced non-traditional machining process widely employed for precision machining of electrically conductive materials, particularly those that are hard, brittle, or difficult to machine using conventional methods [1]. In WEDM, material removal occurs through a series of controlled electrical discharges between a continuously moving thin wire electrode and the workpiece, both submerged in a dielectric medium. The intense

localized heat generated by these electrical sparks results in melting and vaporization of the workpiece material, enabling the fabrication of complex geometries with high dimensional accuracy and excellent surface finish [2].

The increasing demand for miniaturized, high-precision components in aerospace, automotive, medical, and tool-manufacturing industries has driven the adoption of advanced machining technologies. Conventional machining processes such as turning, milling, and grinding often encounter limitations when machining high-strength and corrosion-resistant alloys due to excessive tool wear, work hardening, and poor surface quality. WEDM overcomes these limitations by eliminating mechanical contact between the cutting tool and the workpiece, thereby reducing cutting forces, tool wear, and residual stresses while maintaining high machining accuracy [3, 4].

Stainless steel SS304 is one of the most commonly used austenitic stainless steels because of its excellent corrosion resistance, toughness, good mechanical properties, and biocompatibility. It is extensively used in chemical processing equipment, aerospace components, medical instruments, and precision gauges. However, SS304 exhibits poor machinability under conventional machining conditions due to its tendency to work-harden and its low thermal conductivity. As a result, WEDM has emerged as an effective and reliable alternative for machining SS304 components with intricate profiles and strict tolerance requirements [5, 6].

Despite its widespread industrial application, achieving consistent machining quality in WEDM remains a challenge because of the complex interaction among numerous process parameters such as pulse on-time, pulse off-time, peak current, servo voltage, and wire-related parameters. Improper selection of these parameters may lead to poor surface finish, dimensional inaccuracies, wire breakage, and reduced productivity. Therefore, systematic investigation and optimization of WEDM process parameters are essential to enhance machining performance and surface integrity [7-9].

2. Working Principle of WEDM

Wire Electrical Discharge Machining operates on the principle of electro-thermal erosion, where material is removed from the workpiece by a series of discrete electrical discharges. In this process, a thin metallic wire commonly made of brass or copper acts as the electrode and is continuously fed between two wire guides while maintaining a small spark gap from the workpiece. Both the wire and the workpiece are immersed in a dielectric fluid, typically deionized water, which acts as an insulating medium until electrical breakdown occurs [10].

When a pulsed voltage is applied between the wire electrode and the workpiece, the dielectric medium ionizes, forming a plasma channel. This results in the generation of a high-energy spark with temperatures ranging from approximately 8000 to 12,000 °C. The intense heat causes localized melting and vaporization of the workpiece material at the point of discharge. The molten

material is rapidly cooled and flushed away by the circulating dielectric fluid, leaving behind microscopic craters on the machined surface [11].

The wire electrode continuously moves during machining to prevent excessive wear and maintain a consistent cutting action. The relative motion between the wire and the workpiece is precisely controlled by a CNC system, allowing accurate machining of complex two-dimensional and three-dimensional contours. The machining performance and surface quality are strongly influenced by electrical parameters such as pulse on-time, pulse off-time, and peak current, as well as by wire tension, wire feed rate, and dielectric flushing conditions [12].

Overall, the non-contact nature of WEDM enables the machining of delicate and hard materials without inducing mechanical stresses, making it highly suitable for precision manufacturing applications requiring high accuracy and superior surface integrity [13].

3. Experimental Methodology

The experimental methodology adopted in the present study was designed to systematically investigate the effect of selected Wire Electrical Discharge Machining (WEDM) process parameters on surface roughness during machining of SS304 stainless steel. A structured approach involving selection of materials, machining parameters, experimental design, and measurement techniques was followed to ensure accuracy and repeatability of results [14].

3.1 Workpiece Material

SS304 stainless steel was selected as the workpiece material due to its widespread industrial applications and machining challenges under conventional processes. The chemical composition and mechanical properties of SS304 make it suitable for evaluating the effectiveness of WEDM. The workpieces were prepared in rectangular blocks of uniform dimensions to ensure consistency across all experimental trials. Prior to machining, the specimens were cleaned to remove surface contaminants [15].



Figure 1: SS304 Steel

Table 2. Chemical Composition of SS304 Steel Alloys

Compound	Wt%	Compound	Wt%
Cr	12.6	Mn	1.0
C	0.15	Ni	0.06
Si	1.0	S	0.02
Mg	0.98	Fe	Bal.
P	0.03		

3.2 Electrode Wire and Dielectric Fluid

A brass wire electrode of standard diameter (commonly 0.25 mm) was used for all experiments because of its good electrical conductivity, thermal stability, and widespread industrial usage [16]. The wire was continuously fed during machining to maintain stable discharge conditions and minimize wire wear.

Deionized water was used as the dielectric fluid. It served the dual purpose of insulating the electrode and workpiece until electrical breakdown occurred and flushing away molten material and debris from the spark gap. Proper filtration and circulation of the dielectric were maintained throughout the experiments.

3.3 WEDM Machine Tool

All experiments were conducted on a CNC-controlled Wire Electrical Discharge Machining machine. The machine was equipped with a servo-controlled power supply capable of independently varying pulse on-time, pulse off-time, and peak current. CNC programming was employed to ensure accurate wire movement and repeatability of the machining path for each experimental run [17].

3.4 Selection of Process Parameters

Based on an extensive literature review and preliminary trials, three key WEDM process parameters were selected as control factors:

1. **Pulse on-time (Ton):** Duration for which electrical discharge is applied
2. **Pulse off-time (Toff):** Time interval between successive discharges
3. **Peak current (Ip):** Maximum current during discharge

These parameters were varied at predefined levels while keeping other parameters such as wire tension, wire feed rate, servo voltage, and flushing pressure constant to isolate their effects on surface roughness [18].

3.5 Design of Experiments

Taguchi's design of experiments (DOE) technique was employed to minimize the number of experimental trials while effectively analyzing the influence of machining parameters. An appropriate orthogonal array was selected based on the number of control factors and their levels. This approach enabled systematic evaluation of parameter effects with reduced experimental cost and time [19].

The "smaller-is-better" signal-to-noise (S/N) ratio criterion was used for surface roughness optimization, as lower surface roughness values are desirable for improved machining quality.

Table 2 Standards L9 Orthogonal Array

Sr. No	Pulse on Time	Pulse off Time	Current
1	18	2	35
2	18	3	40

3	18	4	45
4	20	2	40
5	20	3	45
6	20	4	35
7	22	2	45
8	22	3	35
9	22	4	40

3.6 Measurement of Surface Roughness

Surface roughness (Ra) was selected as the primary response parameter. After machining, surface roughness measurements were taken using a calibrated surface roughness tester. Multiple readings were recorded at different locations on each machined surface, and the average value was used for analysis to reduce measurement uncertainty.



Fig. 2 Portable Digital Surface Tester TS00 of HOMMEL make.

3.7 Data Analysis and Optimization

The experimental results were analyzed using statistical tools. Signal-to-noise ratio analysis was used to identify optimal parameter levels. Analysis of Variance (ANOVA) was performed to determine the statistical significance and percentage contribution of each process parameter on surface roughness. Optimization was carried out to obtain the parameter combination that yielded minimum surface roughness.

Confirmation experiments were conducted under optimal conditions to validate the predicted results and assess the reliability of the optimization technique.

4. PERFORMANCE ANALYSIS

In this chapter, we are discussing about the effects or influence of machining parameter, i.e. Discharge current, Pulse on time and Current on surface roughness (SR) of SS 410 machined work piece with tungsten carbide tool and find out which parameter is most important during an experiment with the help of Taguchi design.

4.1 Model Analysis for SR

Table 3 Cumulative data for SN Ratio SR

1	18	2	35	2.98	9.4843
2	18	3	40	3.17	10.0212
3	18	4	45	3.47	10.8066
4	20	2	40	4.65	13.3491
5	20	3	45	4.93	13.8569
6	20	4	35	3.99	12.0195
7	22	2	45	5.12	14.1855
8	22	3	35	4.29	12.6491
9	22	4	40	4.56	13.1793

The Table 4.2 and shows the L₉ orthogonal array with repeat measurement of responses for runs one to nine. Repeats of response measurement technique is used overcome the drawback of saturated design in MINITAB software. It also shows that the SN ratio for run one and ten are same as it is calculated for the repeat's measurement. The SN ratio values are calculated with help of MINITAB 17 software

4.2 Main Effects of SR

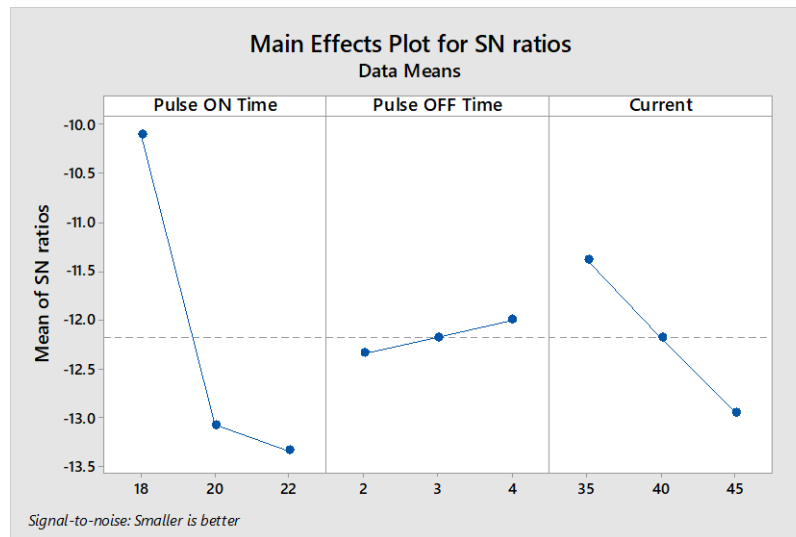


Figure 3 Main effect plots for mean of SN ratio mean of SR

The influence of each control factor (pulse on time, pulse off time, and peak current) Surface roughness was analyzed from the S/N ratio response table, which expresses the S/N ratio at each level of control factor. The control factor influence is determined by its level difference values. A bigger control factor level difference means a greater influence on the SR Calculation of SN ratio: A higher SN ratio with better characteristics was used and was calculated with the help of the MINITAB software for the experimental tests. The values of the complete SN ratio are SN ratio

for the first test. From the graph, it is observed that the optimum material removal rate was in the lowest values of the in the response graph. The optimal input parameters were pulse on time 18sec (level 1), 35 A Current (level 1) and 4 sec pulse off time (level 3). The graph graphically shows the effect of the control factors on SS304 material. The configuration of the process parameters with the highest ratio always provides the optimum quality with a minimum variation. The graph shows the relationship change when the control factor configuration was changed from one level to another.

From the graph of main effects for the media it is observed that the wire electro discharge machining is a function of the pulse on time, pulse off time, Current of the material. Pulse on time is defined as the time during which the machining is performed. The machining process becomes faster after increasing the pulse on time. By increasing the pulse on time the Surface roughness increases. High pulse on time more material gets melted at the tool work piece interface, so there is not enough time for the proper flushing time but as the value of pulse off time is too short flushing to clear the debris from the inter-electrode gap between the tool and work piece, so arcing take place which result in Increase in the IP value will increase the pulse discharge energy which in turn can improve the cutting rate further. With increase in the value of peak current MRR, Ra and WWR increases. Peak current is found to be the major factor affecting the Surface Roughness, material types and strongly depends on the constant machining parameter selected for EDM machine.

These Taguchi results compared to the regression analysis show close agreement. The main effect diagrams for parameters such as pulse on time, pulse off time, current are obtained from the regression analysis using the Minitab Version 17 statistical software to predict the optimal level of measured response.

4.3 Analysis of Variance

The relevance of the models is tested by analysis of variance (ANOVA). It is a statistical tool for testing the null hypothesis for planned experiments, in which several different variables are studied simultaneously. ANOVA is used to quickly analyze the variances in the experiment using the Fisher test (F test). Table shows the result of the ANOVA analysis. The ANOVA analysis makes it possible to observe that the value of P is less than 0.05 in the three parametric sources. It is therefore clear that (1) the pulse on time, (2) the pulse off time, (3) the Current of the material have

an influence on the SS304 steel material. The last column of cumulative anova shows the percentage of each factor in the total variance that indicates the degree of impact on the outcome. Table shows the ANOVA of SS304 Steel & copper wire. The table shows that the pulse on time (59.45%), the Current (17.78 %) and the pulse off time (22.68 %) have a major influence on the surface roughness.

Table 4 ANOVA Result of SR

Source	DF	Adj SS	Adj MS	F-Value	P-Value	% Contribution
T-ON	2	2.85389	1.42695	612.81	0.002	59.45
T-OFF	2	1.08882	0.5444	314.12	0.056	22.68
Current	2	0.85129	0.42564	135.36	0.007	17.78
Error	2	0.00629	0.00314			
Total	8	4.80029				

4.4 Multiple Linear Regression Models

To establish the correlations between the parameters are (1) pulse on time, (2) pulse off time, (3) Current, of SS304 Steel material The model of multiple linear regression was obtained using the statistical software "MINITAB R 17". Terms that are statistically significant are included in the model. The final equation obtained is the following.

Mathematical models of peak voltage, pulse off time and pulse ON time is calculated using MINITAB 17 software and analysis of regression carried out to obtain the predicted value of material removal rate.

Regression Equation

$$SR = -5.77 + 0.3625 \text{ Pulse ON Time} - 0.122 \text{ Pulse OFF Time} + 0.0753 \text{ Current}$$

Substituting the registered values of the variables for the previous equations, It is observed from above regression Equation that the Pulse on time is more effective in the material Surface roughness of the SS304 steel. The important factor that affects Surface roughness is the Pulse on time and the coefficient associated with it is positive. This suggests that the Pulse on time increases surface roughness.

Taguchi design and ANOVA give mathematical model which predict the result nearly accurate. This can be due to this project deals the parameter level in very short range. Also, every effort has done to maintain noise factors constant to greater level throughout experimentation.

4.5 Comparison of Experimental and Mathematical Analysis of SR

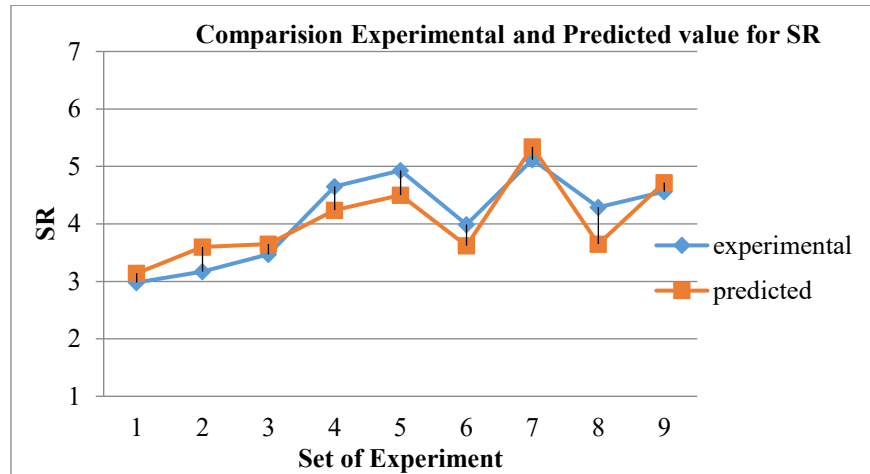


Figure 4 Experimental Vs Predicted Surface Roughness

Table 4 Comparison of Experimental and Predicted Surface Roughness

Sr. No.	Experimental value	Predicted value	Error
1	2.98	3.14	5.09%
2	3.17	3.40	6.76%
3	3.47	3.65	4.93
4	4.65	4.24	9.66
5	4.93	4.50	9.55
6	3.99	3.65	9.31
7	5.12	5.34	4.11
8	4.29	4.47	4.02
9	4.56	4.72	2.01

Shows the comparison of SR results from the mathematical model developed in the present work with values obtained experimentally respectively. By comparing the results of values obtained from SR equation and results of confirmation test, we have concluded that the confirmation test holds the equation for the values which are not included in the orthogonal array. Therefore, the multiple regression equation derived above correlate the evaluation of SR in the composite with the degree of approximation.

Surface roughness decreases for first sample and increases for ninth sample because of Experiment from 1-9 are the set of experiments from Taguchi design L9 array, every experiment has different level of parameters and having unique combination of parameter level. Also

experiment no. 1 having low level of pulse on time, so surface roughness (Ra) having low value i.e better surface finish, as pulse on time having higher contribution relative to other factor.

Table 5: Optimum level of parameters

Sr. No.	Parameter	Optimum level
1	Pulse on time (Level 1)	18
2	Pulse off time (Level 3)	4
4	Peak Current (Level 1)	35

4.7: Confirmation experiment result

Experiments was conducted for T-on at level 3, Current at level 1 and T-off at level 1

Table 6: Confirmation experiment result

Parameter	Model value	Experimental value	Error %
SR	2.90	2.76	4.82

5. CONCLUSIONS

This study covers the observations about the surface roughness over SS304 steel material by the process of WEDM machine for the different input parameters to thoroughly study over the effect of EDM machining process on the SS304 steel material. Throughout the experimentation I got some results as under.

- : This study covers the observations about the Material surface roughness over SS304 steel material by the process of WEDM machine for the different input parameters to thoroughly study over the effect of WEDM machining process. Throughout the experimentation we got some results as under,
- : The optimal solution obtained for SR based on the combination of WEDM parameters and their levels is (i.e. Pulse ON Time 18 μ sec at level 1, Current 35A at level 1 and Pulse OFF Time 4 μ sec at level 3). The Pulse on Time and pulse on time are more significant Machining Parameters than Pulse off time.
- : ANOVA results indicate that pulse on time plays prominent role in determining the surface roughness. The contribution of pulse on time, current and pulse off time to the quality characteristics surface roughness Ra is 59.45%, 22.68% and 17.78% respectively.
- : SR increases with Pulse on Time and Current as they provide high discharge energy.
- : SR decreases with increases Pulse off Time.

- : The optimal cutting parameters are determined using Taguchi methods match with the experimental values by minimum errors i.e.4.82% for SR.
- : Through the developed mathematical models, any experimental results of surface roughness with any combination of laser cutting parameters can be estimated.

Conflict of Interest

The author(s) declare that there is **no conflict of interest** associated with this research work

References

1. Anwar, H., Shather, S., & Khudhir, W. (2024). Multi Responses Optimization of Wire Electrical Discharge Machining (WEDM) Parameters of Tool Steel Using Grey Relation Analysis. *Salud, Ciencia y Tecnología - Serie de Conferencias*. <https://doi.org/10.56294/sctconf2024861>
2. Ugrasen, G., Singh, M., & Ravindra, H. (2018). Optimization of Process Parameters for SS304 in Wire Electrical Discharge Machining using Taguchi's Technique. *Materials Today: Proceedings*, 5, 2877-2883. <https://doi.org/10.1016/j.matpr.2018.01.080>
3. Natarajan, K., Ramakrishnan, H., Gacem, A., Vijayan, V., Karthiga, K., Ali, H., Prakash, B., & Mekonnen, A. (2022). Study on Optimization of WEDM Process Parameters on Stainless Steel. *Journal of Nanomaterials*. <https://doi.org/10.1155/2022/6765721>
4. Somavanshi, P. D., & Sathe, Y. U. (2024). Exploring the synthesis processes of graphene oxide and silicon dioxide nanospheres: A comprehensive study. *Journal of the Textile Association*, 85(4), 416–421. <https://doi.org/10.5281/zenodo>
5. Radhakrishnan, K., & Kesavalu, R. (2025). Investigation of Wire-Cut EDM parameters for machining 2304 duplex stainless steel: effects on material removal rate, surface roughness, and tool wear. *Matéria (Rio de Janeiro)*. <https://doi.org/10.1590/1517-7076-rmat-2024-0797>
6. Jayakumar, K., Vigneshwaran, S., Subash, T., Vignesh, S., & Vignesh, R. (2023). Comparative Wire Electrical Discharge Machining Performance Studies on SS304 and Ti Grade 9 Alloys. *Journal of Materials Engineering and Performance*. <https://doi.org/10.1007/s11665-023-08891-5>
7. Ishfaq, K., Ahmad, N., Jawad, M., Ali, M., & Al-Ahmari, A. (2019). Evaluating Material's Interaction in Wire Electrical Discharge Machining of Stainless Steel (304) for Simultaneous Optimization of Conflicting Responses. *Materials*, 12. <https://doi.org/10.3390/ma12121940>
8. Suresh, T., Jayakumar, K., Selvakumar, G., & Ramprakash, S. (2022). Experimental Investigation on Improvement of Machinability of SS 304 Through Multipass Cutting in WEDM. *Arabian Journal for Science and Engineering*, 48, 11577-11590. <https://doi.org/10.1007/s13369-022-07508-8>

9. Deepthi, N., Keerthi, N., Nagamani, K., Saleemuddin, S., & Kumar, D. (2024). Optimization of Process Parameters In Wire-Edm for Stainless steel -304 By Using Grey-Relational Analysis. *Journal of Physics: Conference Series*, 2837. <https://doi.org/10.1088/1742-6596/2837/1/012071>
10. Seshaiyah, S., Sampathkumar, D., Mariappan, M., Mohankumar, A., Balachandran, G., Kaliyamoorthy, M., Rajendran, B., & Gopal, R. (2022). Optimization on Material Removal Rate and Surface Roughness of Stainless Steel 304 Wire Cut EDM by Response Surface Methodology. *Advances in Materials Science and Engineering*. <https://doi.org/10.1155/2022/6022550>
11. Azawqari, A., Amrani, M., Hezam, L., Baggash, M., & Abidin, Z. (2024). Multi-objectives optimization of WEDM parameters on machining of AISI 304 based on Taguchi method. *The International Journal of Advanced Manufacturing Technology*, 134, 5493 - 5510. <https://doi.org/10.1007/s00170-024-14423-9>
12. Jayakumar, K., & Suresh, T. (2022). Effect of wire materials on performance during WEDM of SS304. *Materials Today: Proceedings*. <https://doi.org/10.1016/j.matpr.2022.03.617>
13. Aboueleaz, M., Naeim, N., Abdelgalil, I., Aly, M., & Elkaseer, A. (2023). Wire Electrical Discharge Machining of AISI304 and AISI316 Alloys: A Comparative Assessment of Machining Responses, Empirical Modeling and Multi-Objective Optimization. *Journal of Manufacturing and Materials Processing*. <https://doi.org/10.3390/jmmp7060194>
14. Biswas, S. (2023). A Review on the Usage of Different Optimization Techniques for Wire Electro Discharge Machining of Stainless Steels. *INTERANTIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT*. <https://doi.org/10.55041/ijrsrem25488>
15. T, S., K, J., P, L., & Vaishnavan, S. (2024). Experimental investigation of 3D taper profile machining of SS304 using WEDM. *Journal of the Chinese Institute of Engineers*, 47, 160 - 172. <https://doi.org/10.1080/02533839.2023.2298973>
16. El-Bahloul, S. (2024). Modeling of Wire EDM Using an Integrated Approach for Machining of AISI304 Stainless Steel. *Management and Production Engineering Review*. <https://doi.org/10.24425/mper.2024.151130>
17. Mahapatra, S., & Patnaik, A. (2007). Optimization of wire electrical discharge machining (WEDM) process parameters using Taguchi method. *The International Journal of Advanced Manufacturing Technology*, 34, 911-925. <https://doi.org/10.1007/s00170-006-0672-6>
18. Somavanshi, P. D., & Sathe, Y. U. (2023). Evaluating the electrochemical performance of graphite and silicon-based materials in energy storage devices. *REDVET – Revista Electrónica de Veterinaria*, 24(4). DOI: <https://doi.org/10.69980/redvet.v24i4.1277>
19. Mao, R., Sun, Z., Gan, S., Lei, W., Du, Y., & Kong, L. (2025). Optimization of Process Parameters for Wire Electrical Discharge Machining of 9Cr18Mov Based on Grey Relational Analysis. *Processes*. <https://doi.org/10.3390/pr13051547>

